

a certain period of time. To reduce the cost of test, pay-per-use option has come up. Here the payment depends on how you test. For testing a product like Apple iPhone, more features are required and therefore you will have to shell out more. Whereas for a product like USB device, you need to pay much lower than for iPhone. How much you pay depends on two-three parameters: For how much time you test, how many boards you test and how complex are the boards to be tested. The more complex the boards, the more the amount to be shelled out for testing the same number of boards.

Today, most of the manufacturing is done by CEMs. That is, if you design a product today, the boards are manufactured by some CEM which has the full facility to service a hundred customers. CEMs survive on multiple orders. Sometimes there could be big orders, sometimes there could be very low volume of manufacturing. The pay-per-use test system works very well for them. It dramatically reduces their cost of test—while maximising flexibility to test boards in a wide range of sizes and complexities. Let's say, a CEM has a huge order which requires three months of complete activity and then two months of low activity. In these two months, they hardly spend anything on testing. So they purchase a full-capability test system at a much lower price than a conventional system and pay only for the capabilities they need, only when they use them.

In order to increase the efficiency of testing, one of the things that manufacturers have been doing over the years is trying to see whether they should test every component on the board or do selective testing.

Let's say, you have a cellphone which has 15-20 components—one big ASIC, amplifier, receiver, antenna and so on. Ideally, you would like to test every component on the board to ensure that everything is working perfectly.

With functional testing, you need not test every component. You just

test the functions that the cellphone is expected to do. So, typically, functional testing is end-to-end testing where an input is given and output checked. If the whole device is working, it is assumed that all its components are working.

There is also a possibility that critical limits aren't met. So manufacturers do sample testing. In sample testing, some samples are tested for complete test. The samples are tested for every aspect—voltage level, current level, functionality, etc.

If the manufacturing volume is high and you have a very standard parts supplier, you have a certain level of confidence over the supplier that the parts being supplied are certified and wouldn't like to do the testing exercise over all the components. However, you still do sample testing to ensure parts' quality. For a new product launch, obviously every component is tested because you haven't built the confidence yet over the vendor that the components supplied are good.

So for mass manufacturing, there could be a mix of functional testing and complete parametric measurements for sample devices.

To make testing simpler, boundary scan has become an essential tool. It enables much of a board to be tested with minimal access. Over a period of time, JTAG port has become a standard connector on many devices by accessing which you can do many things on the device—you can test the device, program the device and so on. It was introduced primarily to reduce the cost of test. The maximum cost of test in manufacturing is involved in ICT. ICT means every pin of the device has to be accessed for testing. Traditionally, for a 144-pin device, you need to probe every pin, apply signal and check the response. With JTAG, even if it's a 144- or 200-pin device, through only five pins you can check the functionality of the device.

"In today's production process, automation is the key for optimising the cost of the final product. With in-

terfacing of test equipment with PCs and PLCs, and features like pass/fail, these are being integrated within the process chain. These instruments not only measure the parameters but also do data logging for documentation, statistical quality controls, etc. When a component fails to comply with specifications, a solenoid control signal is generated and the faulty part moved out. An example is a high-precision multimeter with pass/fail output used in a resistor, capacitor or inductor component manufacturing unit," adds P. Prabhu, general manager (technical), Scientific Mes-Technik.

According to Jayaram Pillai, managing director, National Instruments IndRA (India, Russia and Arabia), key technology trends that increase test efficiency and reduce test costs include organisational test integration, system software stack, heterogeneous computing and IP to the pin.

Organisational test integration.

Throughout the electronics design and manufacturing industry, test teams are improving integration across the organisation to gain a competitive edge. In the past, validation (the process of testing a product during design to guarantee that it meets feature specifications) and manufacturing test teams had few opportunities to work together. However, test managers seeking to decrease time to market and reduce test costs see that improving. Increased use of automation across both the groups has shown that they can share a common software and instrumentation. Hence one of the top goals for test engineering organisations is to increase reuse between validation and production.

System software stack.

Software has been a critical component of automated test systems since it was first used to control standalone instruments more than 40 years ago. In fact, software development costs are often two to ten times more than capital costs of most test systems today. In response to rising software development costs and accelerated product development cycles, today's industry-leading com-